

Funcția indicator

$$1_A(x) = \begin{cases} 1, & x \in A \\ 0, & x \notin A \end{cases}$$

$$f: A \rightarrow B$$

$$\text{supp } f = \{x \in A \mid f(x) \neq 0\}$$

Obs. Dacă f este densitate de prob. $\text{supp } f = \{x \in \mathbb{R} \mid f(x) > 0\}$

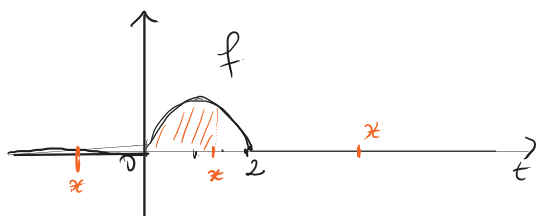
ex $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = \begin{cases} c(4x - 2x^2), & x \in \underline{(0,2)} \\ 0, & \text{altfel} \end{cases} = c \cdot (4x - 2x^2) \cdot 1_{\underline{(0,2)}}(x)$

$\text{supp } f$

$$\int_{-\infty}^{\infty} f(x) dx = 1 \Leftrightarrow \int_{\text{supp } f} f(x) dx = 1$$

$$E(X) = \int_{-\infty}^{\infty} x \cdot f(x) dx = \int_{\text{supp } f} x \cdot f(x) dx$$

$$f(x) = \begin{cases} 0, & x \leq 0 \\ c(4x - 2x^2), & x \in (0,2) \\ 0, & x \geq 2 \end{cases}$$



$$F(x) = \int_{-\infty}^x f(t) dt$$